

Evaluation Artefacts in Physics-Informed Band-Gap Prediction: A Leakage-Controlled Study of Two-Dimensional Materials

ADHVAY JAGADEESH¹, RUTVI MUDALAGI¹, and MARX AKL¹

¹Aspiring Scholars Directed Research Program (ASDRP), Fremont, CA 94539 USA (e-mail: mail.ack.ventures@gmail.com)

Corresponding author: Marx Akl (e-mail: makl@asdrp.org).

This work was conducted through the Aspiring Scholars Directed Research Program (ASDRP), Fremont, CA, USA.

ABSTRACT Hybrid architectures that correct a gradient-boosted prior with a neural residual head are widely reported to improve tabular property prediction in materials informatics. We show that one such improvement, previously reported for band gaps of two-dimensional materials, is an artefact of how the model was evaluated rather than a property of the model. Under a protocol that closes six leakage pathways — identifier-like features, whole-dataset encoding, in-sample construction of stacked priors, polymorphs straddling folds, absent held-out data, and significance tests applied per sample rather than per fold — the hybrid does not improve on the prior it corrects ($R^2 = 0.810 \pm 0.047$ against 0.823 ± 0.043 ; superior on 12 of 25 fold estimates; $p = 0.27$), and no physics-motivated component contributes measurably. Reverting a single implementation detail, constructing the prior in sample rather than out of fold, reproduces the original gain on demand, because a prior that appears near-perfect on training rows trains the corrector to emit nothing. On 1,169 rectangular-lattice C2DB materials a stacked ensemble of five heterogeneous learners is the strongest configuration ($R^2 = 0.880 \pm 0.037$, MAE 0.360 eV). Because its margin over the best single model is small, we justify it on decision-relevant grounds: the best single model is unstable across folds, the ensemble beats a committed choice on 15 of 15 folds, cuts errors above 1 eV from 7.3% to 6.1%, and trains in 8 seconds against the CPU-hours it replaces. We additionally decompose accuracy by descriptor cost, finding that removing every DFT-derived descriptor costs 0.044 in R^2 .

INDEX TERMS Cross-validation, data leakage, ensemble learning, materials informatics, model evaluation, physics-informed neural networks, reproducibility, two-dimensional materials.

I. INTRODUCTION

ACCURATE prediction of electronic band gaps (E_g) is a foundational objective in computational materials design. The magnitude and nature of the band gap govern whether a low-dimensional material acts as a metal, semiconductor, or insulator, and determine candidate suitability for field-effect transistors [1], valleytronic architectures [2], [3], and photocatalytic water splitting [4]. Density functional theory with hybrid functionals such as HSE06 [5] offers excellent accuracy at computational scaling that makes exhaustive screening impractical, which motivates fast surrogate estimators.

For heterogeneous tabular materials data, decision-tree ensembles [6], [7] frequently match or exceed deep architectures [8], [9], though the conditions under which this holds are narrower than often assumed [10]. A natural response is the hybrid residual design: a tree supplies a prior estimate and

a neural network learns a correction, optionally regularised by physical constraints in the manner of physics-informed neural networks [11], [12]. Such designs are attractive because they appear to combine the sample efficiency of ensembles with the smoothness of neural function approximation, and because anisotropic two-dimensional systems — where in-plane lattice vectors are unequal and strain couples strongly to the electronic structure [13]–[15] — are precisely the setting in which axis-aligned tree partitions appear least suited.

An earlier version of this work reported exactly that: a gradient-boosted prior coupled to a quantile residual head with an engineered structural layer, improving pooled R^2 from 0.7323 to 0.7431 with a nominally significant p -value. This article reports what happened when that pipeline was re-evaluated under controls for data leakage. The improvement did not survive, and the reason turns out to be a single line of the training procedure rather than anything about the architec-

ture. Leakage of the kinds we identify has been argued to be a principal driver of irreproducibility across machine-learning-based science [16], and the specific failure we document is easy to introduce by accident and difficult to detect from reported results alone.

Our contributions are four.

- 1) **A leakage-controlled evaluation protocol** for tabular materials regression, closing six specific pathways: identifier-like features, whole-dataset encoding, in-sample construction of stacked priors, polymorphs straddling folds, absent held-out data, and significance tests applied at the sample rather than the fold level.
- 2) **A negative result with an identified mechanism.** Under that protocol the physics-informed hybrid does not improve on its own prior, and no physics-motivated component contributes measurably. We show that the earlier apparent gain arises because an in-sample prior trains the corrector to be inert, and we reproduce it deliberately by reverting that one detail.
- 3) **A decision-relevant justification for ensembling.** Where a small accuracy margin cannot carry the argument, we quantify model-selection risk, large-error rates, and measured wall-clock cost instead.
- 4) **A decomposition of accuracy by descriptor cost,** separating what composition and symmetry provide from what a relaxed geometry and a converged DFT calculation add.

II. METHODS

A. DATASET AND TARGET

We use 1,169 rectangular-lattice (*op*) two-dimensional monolayers from the Computational 2D Materials Database [17] carrying an HSE06 band gap [5]. The target $y \in \mathbb{R}^+$ spans 0.001–8.85 eV with mean 2.60 eV.

An earlier version of this work reported 930 materials. That figure resulted from a blanket missing-value filter applied across all columns, which discarded 239 labelled entries because one descriptor — total magnetic moment — was absent. We impute that column from training-fold medians with an explicit missingness indicator, recovering the full labelled set.

B. DESCRIPTOR TIERS BY DFT COST

C2DB descriptors are not equally cheap, and a claim to avoid expensive computation is only meaningful if it is stated against a defined cost boundary. We therefore partition them (Table 1) and evaluate every tier.

The packing-density proxy is

$$\rho_{\text{cell}} = \frac{N_{\text{atoms}}}{A_{\text{cell}}}, \quad (1)$$

where N_{atoms} is the atom count of the formula unit and A_{cell} the unit-cell area. The DFT-free tier is obtainable from a chemical formula and a candidate space group alone. No tier contains a band gap at any level of theory, which distinguishes this setting from Δ -learning studies that supply a semi-local gap as an input [20].

TABLE 1. Descriptor tiers, by what must be computed to obtain them.

Tier	Descriptors
DFT-free	120 Magpie elemental-property statistics [18] via matminer [19], atom count, layer group, stoichiometry prototype
+ Geometry	Above, plus thickness, unit-cell area, and packing density (Eq. (1)); requires a relaxed geometry
+ Energies (full)	Above, plus energy above hull, heat of formation, total energy, vacuum level, magnetic moment, magnetic state (requires converged DFT)

Magpie statistics replace the raw chemical formula. An earlier pipeline one-hot encoded the formula string into 1,077 indicator columns, of which 825 occurred in exactly one material; such columns are row identifiers rather than descriptors, and a formula appearing only in a validation fold contributes an all-zero block.

C. EVALUATION PROTOCOL

Composition grouping. 176 labelled materials share a formula with at least one other entry. These are polymorphs, distinguished by layer group, whose gaps differ by up to 2.92 eV. Because composition-derived descriptors are near-identical within such a pair, an ungrouped split lets a model memorise one polymorph and be scored on another. All splits are grouped on chemical formula.

Fold-local preprocessing. Category vocabularies, imputation medians, and standardisation statistics are estimated on the training partition only. For a feature j and training fold $\mathcal{T}$,

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j^{(\mathcal{T})}}{\sigma_j^{(\mathcal{T})}}, \quad (2)$$

with $\mu_j^{(\mathcal{T})}$ and $\sigma_j^{(\mathcal{T})}$ computed over $\mathcal{T}$ alone and applied unchanged to the held-out partition.

Out-of-fold stacked inputs. Meta-learners and residual heads are trained against out-of-fold predictions, never in-sample predictions of a fitted base model [21]. Partitioning the training fold into K inner blocks $\{\mathcal{I}_1, \dots, \mathcal{I}_K\}$, the out-of-fold prediction of base learner m for sample $i \in \mathcal{I}_k$ is

$$z_i^{(m)} = f_{-\mathcal{I}_k}^{(m)}(\mathbf{x}_i), \quad (3)$$

where $f_{-\mathcal{I}_k}^{(m)}$ is model m fitted with block $\mathcal{I}_k$ withheld. Section III-C shows that replacing Eq. (3) with the in-sample alternative $f^{(m)}(\mathbf{x}_i)$ is sufficient on its own to manufacture the improvement reported in earlier work.

Held-out evaluation. We report eight independent formula-grouped held-out sets (≈ 234 materials each) in addition to cross-validation, since a single split of this size cannot resolve closely matched models [22].

Corrected inference. We report five repeats of grouped five-fold cross-validation and compare models with the Nadeau–Bengio variance-corrected paired t -test [23]. For

per-fold score differences $d_1, \dots, d_n$ between two models, the naive paired statistic understates variance because the n training partitions overlap. The corrected statistic inflates the variance by the ratio of test to training set size,

$$t_{\text{NB}} = \frac{\bar{d}}{\sqrt{\left(\frac{1}{n} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_d^2}}, \quad (4)$$

where $\bar{d}$ and s_d^2 are the sample mean and variance of the differences, and t_{NB} is referred to t_{n-1} . Every p -value we report uses Eq. (4). An earlier revision applied a paired test to 930 per-sample squared errors, treating non-independent quantities as independent observations of model performance.

D. MODELS

Base learners are random forest, gradient boosting, support vector regression [24], a deep multilayer perceptron with SiLU activations [25], [26] and batch normalisation [27], and a physics-informed network sharing the MLP backbone with a boundary penalty

$$\mathcal{L}_{\text{phy}} = \text{ReLU}(-\hat{y}) + 0.1 \left(e^{\text{ReLU}(0.05-\hat{y})} - 1 \right) \quad (5)$$

discouraging physically impossible negative gaps. Two meta-learners combine their out-of-fold predictions $\mathbf{z}_i = [z_i^{(1)}, \dots, z_i^{(M)}]^\top$ from Eq. (3). The ridge meta-learner solves

$$\hat{\beta} = \arg \min_{\beta} \sum_{i \in \mathcal{T}} \left(y_i - \beta^\top \mathbf{z}_i \right)^2 + \lambda \|\beta\|_2^2, \quad (6)$$

with λ selected by generalised cross-validation within the training fold. The non-negative variant [28] additionally constrains $\beta_m \geq 0$, giving weights that are directly readable as ensemble shares and a prediction confined to the convex cone of its members:

$$\hat{y}_i = \sum_{m=1}^M \beta_m z_i^{(m)}, \quad \beta_m \geq 0. \quad (7)$$

The hybrid forms $\mathbf{x}_{\text{hybrid}} = [\mathbf{x}, \mathbf{x} \odot \hat{\mathbf{y}}_{\text{prior}}, \hat{\mathbf{y}}_{\text{prior}}]$ and adds a learned correction to the prior, with a structural coupling layer building

$$\mathbf{s} = \left[g_1 g_2, g_1^2 - g_2^2, \sqrt{g_1^2 + g_2^2 + \epsilon}, g_1, g_2 \right] \quad (8)$$

from two geometric channels g_1, g_2 , projecting it back onto the feature map through a small multilayer block. Three heads predict residual quantiles at $\tau \in \{0.25, 0.50, 0.75\}$ under the pinball loss [29]

$$\mathcal{L}_\tau = \frac{1}{N} \sum_{i=1}^N \max(\tau(y_i - q_{\tau,i}), (\tau - 1)(y_i - q_{\tau,i})), \quad (9)$$

alongside a point residual head $r(\cdot)$. The correction is combined with the prior as

$$\hat{\mathbf{y}}_{\text{hybrid}} = \hat{\mathbf{y}}_{\text{prior}} + \frac{3}{4} \mathbf{r} + \frac{1}{4} [q_{0.50} + 0.1 \sigma(\Delta q) \Delta q], \quad (10)$$

TABLE 2. Stacked-ensemble accuracy by descriptor tier, 15 fold estimates.

Tier	R^2	MAE [eV]	ΔR^2
All descriptors	0.880 ± 0.037	0.360	—
No DFT energies	0.845 ± 0.046	0.398	−0.035
No DFT at all	0.836 ± 0.057	0.417	−0.044
<i>Prior published hybrid, all descriptors: $R^2 = 0.743$, MAE = 0.555</i>			

where $\Delta q = q_{0.75} - q_{0.25}$ and $\sigma(\cdot)$ is the logistic function. One correction to the earlier implementation bears on every physical claim made about this layer: the channels were selected by array position (indices 0 and 1), which in the assembled matrix were energy above hull and heat of formation — two formation energies, not geometric quantities. Channels are now resolved by name. The available C2DB export contains no in-plane lattice vectors a and b , so we describe Eq. (8) as a soft geometric inductive bias and make no claim that it derives from elasticity theory.

For an unbiased comparison against prior work we additionally reproduce, at their original hyperparameters, the three estimators specified by our earlier pipeline: XGBoost (500 trees, depth 5, $\eta = 0.03$), gradient boosting (400 trees, depth 4), and random forest (400 trees, depth 14).

III. RESULTS

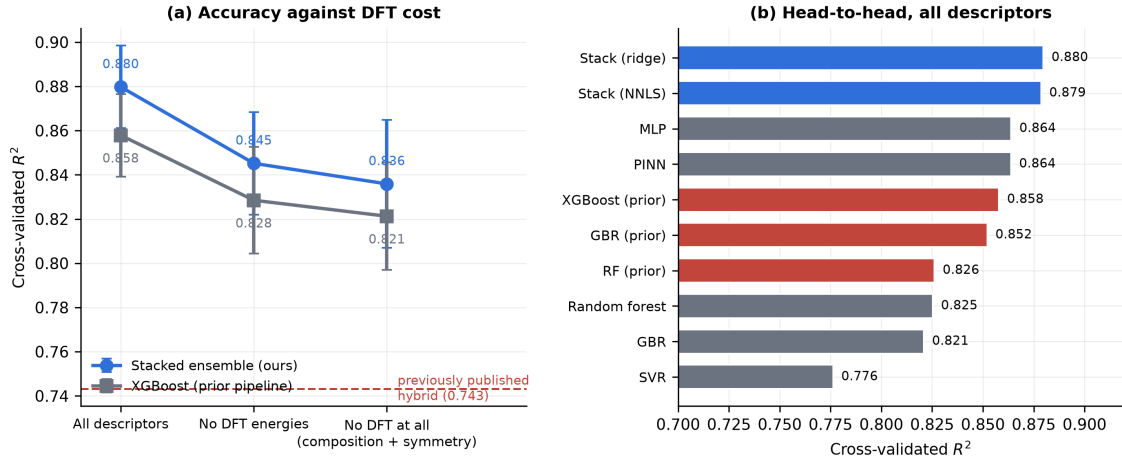
A. MARGINAL VALUE OF DFT-DERIVED DESCRIPTORS

Table 2 and Fig. 1 report accuracy in each descriptor tier.

Removing every DFT-derived descriptor costs 0.044 in R^2 and 0.057 eV in MAE. Most of the attainable accuracy therefore rests on composition and symmetry, and the six DFT energy channels together buy less than a tenth of the variance explained.

We do not present this as a novel capability. Composition-based band-gap models are well established for inorganic solids [30]–[32], and composition-derived descriptors have already been applied to C2DB monolayers specifically: a recent transfer-learning study built 290 compositional descriptors over 2,915 monolayers and reports MAE 0.27 eV [33], better than the 0.417 eV of our DFT-free tier on a smaller and structurally narrower subset. Restricting attention to rectangular-lattice materials narrows the sample; it does not by itself make the composition-to-gap mapping a different problem.

What the decomposition adds is the marginal accounting itself. Studies of composition-only models typically report a single accuracy figure for one fixed descriptor set; separating the tiers tells a practitioner which calculations are worth running before screening, and establishes the baseline against which the hybrid in Section III-C must be judged. It also interacts with a known weakness of composition-based models. Composition alone cannot distinguish polymorphs of one stoichiometry [31], and 176 of our materials share a formula with at least one other entry, differing in gap by up to 2.92 eV. Our DFT-free tier pairs composition with a symmetry label, and our folds are grouped on formula, so no



Red bars are the estimators specified by the prior pipeline, at their original hyperparameters, on identical folds. Removing every DFT-derived descriptor costs 0.044 in R^2 and still exceeds the published hybrid.

FIGURE 1. (a) Accuracy against DFT cost. Removing every DFT-derived descriptor costs 0.044 in R^2 , and the resulting model still exceeds the previously published full-descriptor hybrid. (b) Head-to-head within the full tier; red bars are the estimators specified by the prior pipeline, at their original hyperparameters, on identical folds.

polymorph is ever scored against a sibling seen in training — a confound the composition-only literature names but which is rarely controlled in cross-validation, and which inflates accuracy when left uncontrolled.

Fig. 2 shows the pooled out-of-fold agreement with the HSE06 reference and the regression error characteristic, which reports the fraction of materials resolved within a given absolute tolerance.

Fig. 1(b) gives the head-to-head against the prior pipeline within the full tier. The ensemble beats XGBoost on 14 of 15 fold estimates ($p = 0.025$), gradient boosting on 14 of 15 ($p = 0.002$), and random forest on 15 of 15 ($p < 0.001$). Within the DFT-free tier the margin over XGBoost narrows and is not significant ($p = 0.097$); we report this as a limitation rather than as support.

Fig. 3 shows that the ensemble’s advantage is not an artefact of sample size: it leads at every training-set size tested, and its margin over the tree baselines widens as data are added, with no sign of saturation at the full 792-material training partition.

B. MODEL COMPARISON

Table 3 reports all models over 25 fold estimates.

Both neural models outperform every tree ensemble. With identifier-like features removed and composition statistics supplied, this task falls outside the regime in which boosted trees dominate neural networks [8]–[10].

Across eight independent held-out sets the ordering is reproduced: stacked ridge 0.879 ± 0.025 , PINN 0.866 ± 0.028 , MLP 0.858 ± 0.032 , GBR 0.823 ± 0.025 . The ensemble beats GBR on 8 of 8 splits ($p = 0.001$) and the MLP on 8 of 8 ($p = 0.041$); its margin over the PINN is positive on 7 of 8 splits but only marginally significant ($p = 0.051$). Per-split results are shown in Fig. 5. We report eight sets rather than one because a single split of this size cannot resolve closely

matched models: on one 179-material split the ensemble and the PINN were indistinguishable (0.856 against 0.855), a null that reflects the resolution of that split rather than equivalence of the models.

C. THE HYBRID DOES NOT IMPROVE ON ITS PRIOR

The hybrid is superior to its own gradient-boosted prior on 12 of 25 fold estimates, and the corrected test does not reject the null ($p = 0.27$). Table 4 reports the component ablation; every configuration listed was trained and evaluated.

No physics-motivated component contributes positively, and the tree prior alone is the strongest configuration in the table (Fig. 6). The second row identifies the mechanism: reverting to an in-sample prior *raises* measured R^2 . With a prior that appears near-perfect on training rows, the correction head has almost no residual signal to fit; it learns to emit approximately zero and degenerates into a pass-through of the tree. The previously reported gain is therefore substantially an artefact of the corrector being trained to be inert, and the ablation recovers that artefact on demand by changing one line.

We also tested the natural remedy. A shrinkage gate constrains the correction by $\sigma(\cdot) \in (0, 1)$, initialised near zero so the model begins as a pass-through and must earn any departure. It did not work: the gate converged small (0.081 ± 0.007) but accuracy destabilised ($R^2 = 0.681 \pm 0.237$, with 7 of 25 fold estimates below 0.70 and one at -0.24). Shrinking the gate rescales the gradient reaching the head, whose raw output grows to compensate, leaving the product poorly controlled on unseen data.

D. UNCERTAINTY CALIBRATION

The hybrid predicts quantiles at $\tau \in \{0.25, 0.50, 0.75\}$, previously described as uncertainty estimates. Empirical cov-

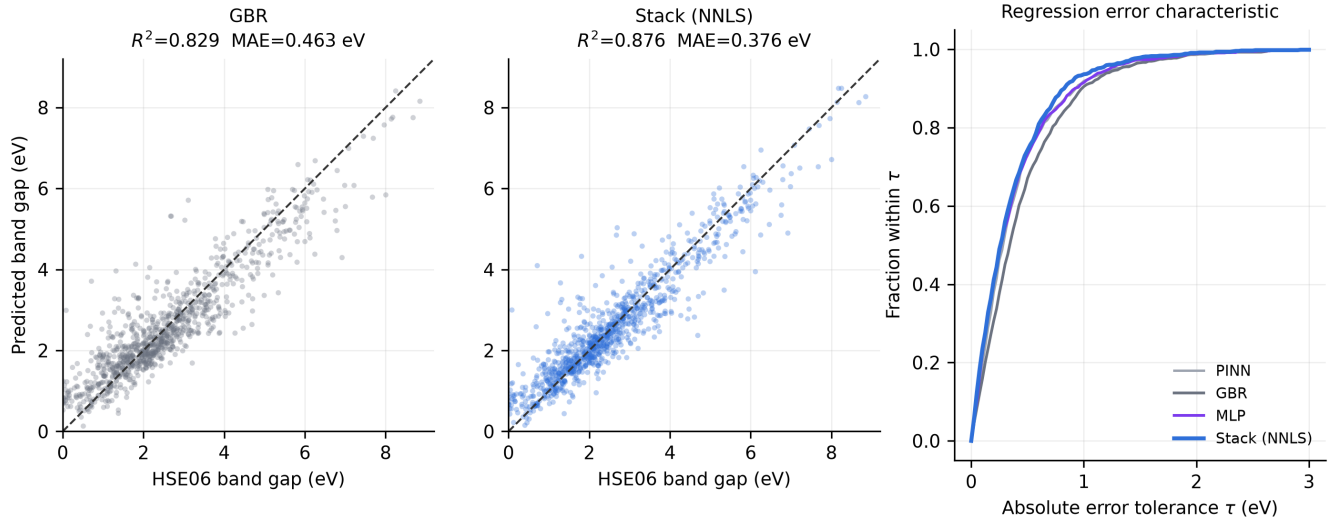


FIGURE 2. Left and centre: pooled out-of-fold predictions against HSE06 reference values for the gradient-boosting baseline and the stacked ensemble. Right: regression error characteristic curves, giving the fraction of materials predicted within a given absolute tolerance τ .

TABLE 3. Repeated grouped five-fold cross-validation, all descriptors, 25 fold estimates. “Folds” counts estimates beating the GBR baseline; p -values are Nadeau–Bengio corrected, one-sided.

Model	R^2	MAE [eV]	Folds	p
Stacked ensemble (ridge)	0.877 ± 0.038	0.363	25/25	< 0.001
Stacked ensemble (NNLS)	0.875 ± 0.038	0.364	25/25	< 0.001
Deep MLP	0.864 ± 0.039	0.378	24/25	0.002
Standalone PINN	0.856 ± 0.043	0.385	22/25	0.033
Random forest	0.827 ± 0.041	0.442	13/25	0.323
GBR (baseline)	0.823 ± 0.043	0.464	—	—
Hybrid GBR + corrector	0.810 ± 0.047	0.445	12/25	0.271
SVR	0.774 ± 0.039	0.497	1/25	—
Hybrid + shrinkage gate	0.681 ± 0.237	0.571	3/25	0.090

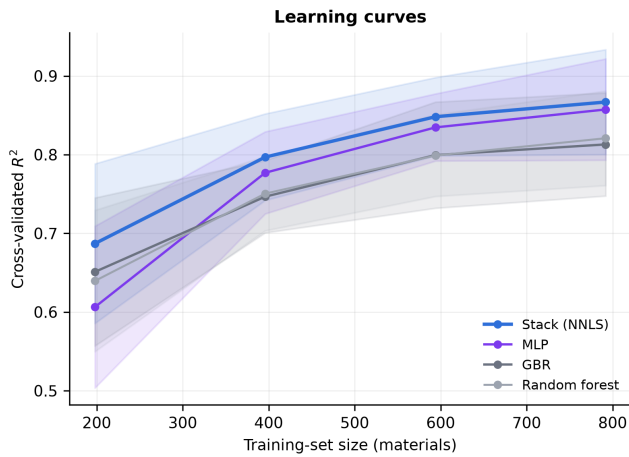


FIGURE 3. Learning curves under grouped cross-validation. Shaded bands are 95% confidence intervals over folds.

erage of the predicted interval,

$$C = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[q_{0.25,i} \leq y_i \leq q_{0.75,i}], \quad (11)$$

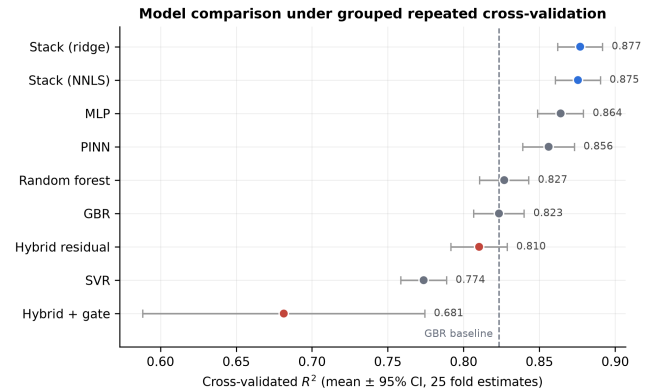


FIGURE 4. Cross-validated R^2 with 95% confidence intervals over 25 fold estimates. Hybrid variants are shown in red.

should equal the nominal level of 0.50; the measured value is $C = 0.172$ at a mean interval width of 2.47 eV (Fig. 7). The intervals are simultaneously wide and badly miscalibrated, and the deficit persists across every nominal level tested (Fig. 7(a)). We therefore withdraw the uncertainty claim; supporting one requires scoring the predictive distribution

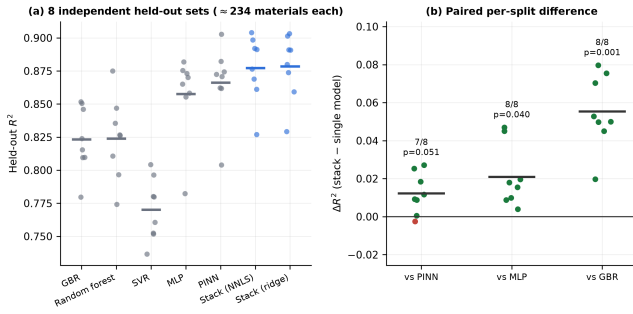


FIGURE 5. Repeated held-out evaluation. (a) Per-split R^2 on eight independent formula-grouped held-out sets; horizontal bars are means. (b) Paired per-split difference between the stacked ensemble and each single model, annotated with splits won and the one-sided p -value.

TABLE 4. Component ablation of the hybrid, pooled out-of-fold R^2 .

Configuration	R^2	Δ vs. full
Tree prior alone, no corrector	0.819	+0.018
In-sample prior (leaky)	0.817	+0.015
Without boundary penalty	0.806	+0.004
Without structural layer	0.804	+0.002
Without anisotropy penalty	0.804	+0.002
Full framework	0.801	—
Without quantile heads	0.796	−0.005
Without residual head	0.766	−0.035

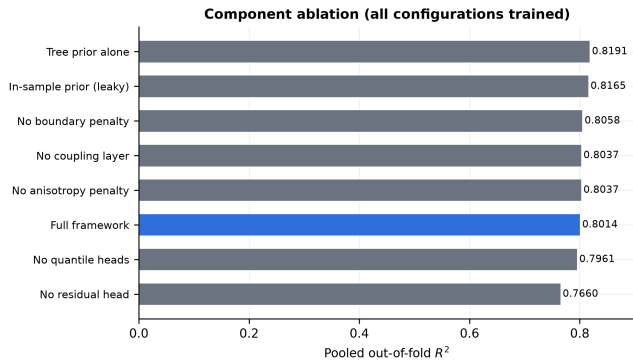


FIGURE 6. Component ablation of the hybrid. Every configuration shown was trained and evaluated; no physics-motivated component contributes positively, and the tree prior alone is the strongest configuration.

with a proper rule [34].

E. WHY THE ENSEMBLE IS WORTH ITS COST

The ensemble's margin over the best single model is small ($\Delta R^2 \approx 0.016$) and would not by itself justify a fivefold training cost. We therefore evaluate it on three decision-relevant axes (Fig. 8).

Model-selection risk. The best single model is not the same model on every fold: the MLP wins 8 of 15 and the PINN 7, and no tree model ever wins. A practitioner must commit to one architecture in advance, and cannot know which. The ensemble beats the best-on-average committed choice on 15 of 15 folds ($p = 0.0001$), and its regret against a per-fold oracle is negative (-0.008) — it outperforms an oracle that

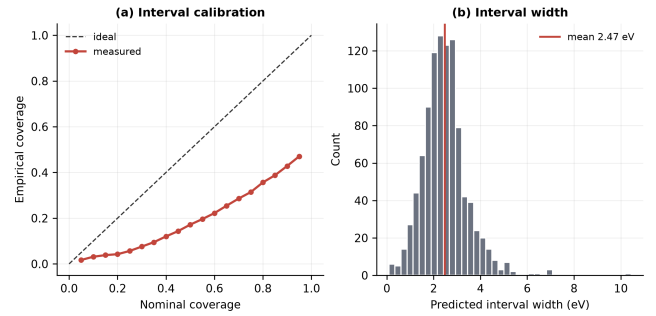


FIGURE 7. Quantile calibration of the hybrid. (a) Empirical against nominal coverage; the model is over-confident at every level. (b) Distribution of predicted interval widths.

always selects the best single model. Committing instead to the weakest plausible choice costs 0.102 in R^2 , roughly six times the ensemble's margin over the strongest.

Large-error rate. Screening is harmed more by rare large errors, which move a material between application classes, than by average error. Predictions in error by more than 1 eV occur at 6.1% for the ensemble against 7.3% for the best single model and 10–13% for the tree baselines; the 95th-percentile absolute error is 1.10 eV against 1.18 eV.

Cost. Training the full ensemble, including the out-of-fold predictions the meta-learner requires, takes 8.0 s against 0.5 s for the MLP alone, and inference over 198 materials takes 22 ms. The relative overhead of $16\times$ is real but the absolute cost is eight seconds, against the CPU-hours of the HSE06 calculation being replaced. The standard objection that an ensemble of N models costs N times more [35] does not bind in this regime.

We report one axis that does not support the ensemble. On a screening task measuring wasted DFT evaluations per true hit, results are mixed: the ensemble is best for a visible-photocatalysis window (1.8–3.0 eV), the PINN is best for a photovoltaic window (1.0–2.0 eV), and several models tie for wide-gap selection.

IV. DISCUSSION

A. WHY RESIDUAL CORRECTION FAILS

Once the prior is constructed out of fold, its residuals on unseen materials are dominated by irreducible noise rather than recoverable structure, and a corrector fitted to them adds variance without removing bias. This is not specific to materials data; residual-boosting correctors are known to yield negative gains when the residual signal is noise-dominated [36]. The practical consequence is that hybrid residual designs must be validated against their own prior with an out-of-fold construction. Compared against an in-sample prior they can appear to help while doing nothing, and that appearance is stable enough to survive review.

B. WHAT THE ENSEMBLE BUYS

The stacking gain arises from error diversity rather than from any member's individual strength. The mean pairwise cor-

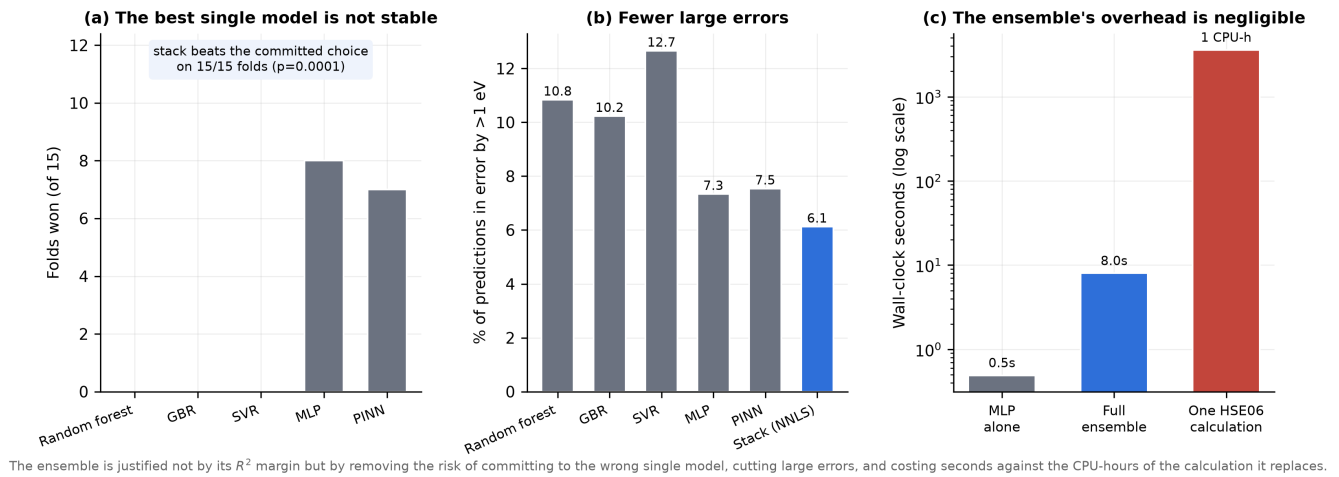


FIGURE 8. (a) The best single model is unstable across folds. (b) Rate of predictions in error by more than 1 eV. (c) Measured training cost against the calculation being replaced, on a logarithmic scale.

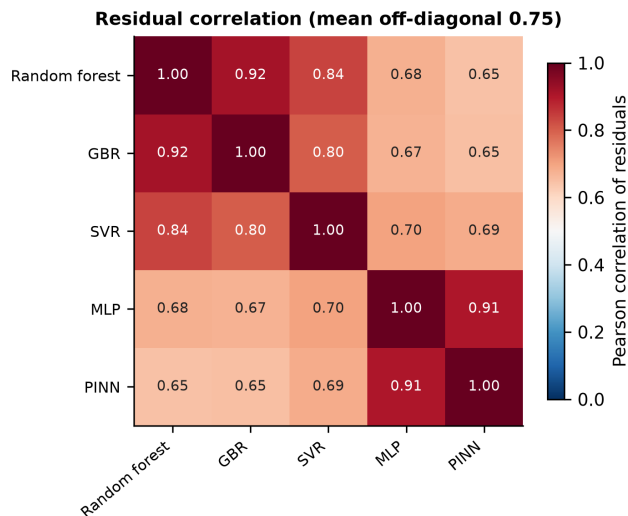


FIGURE 9. Pearson correlation between out-of-fold residuals of the base learners. Lower correlation between clusters is what makes stacking profitable.

relation between base-learner residuals is 0.750, and Fig. 9 shows the neural and tree models forming visibly distinct error clusters; it is combination across those clusters that the meta-learner exploits.

Fig. 10 contrasts the fitted ensemble weights with a drop-one ablation. The two disagree: SVR receives zero weight yet gradient boosting receives 0.238 while contributing nothing measurable when removed. Collinear members split weight between them, so weights alone cannot establish a contribution and the drop-one test is the appropriate instrument. That test orders the members consistently with their individual accuracy, but no single member's contribution is statistically significant once the Nadeau–Bengio correction is applied ($p = 0.11$ – 0.45); the uncorrected test would report $p < 0.05$ for three of them. The defensible statement is that the en-

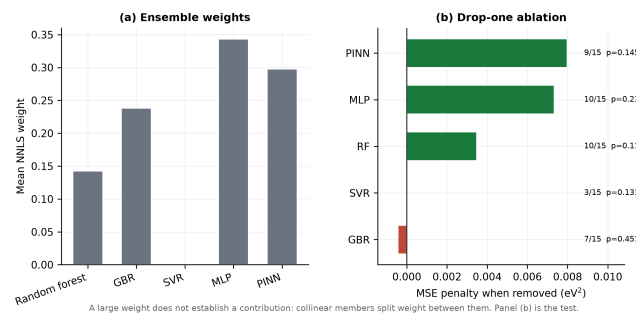


FIGURE 10. Ensemble member contributions. (a) Mean non-negative stacking weights. (b) Drop-one ablation: change in ensemble MSE when each member is removed, with fold estimates degraded and the corrected one-sided p -value. Positive values indicate that removing the member hurts.

semble as a whole outperforms its members, not that any one member is indispensable.

C. ON THE ANISOTROPY HYPOTHESIS

Our results do not refute the physical hypothesis that motivated the original architecture — that in-plane anisotropy shapes band gaps in rectangular 2D lattices [13], [37], [38]. They show that the present implementation cannot test it: the coupling layer was wired to two formation energies, and the available data contain no in-plane lattice vectors. Recovering a and b from a full C2DB structure pull is the single most valuable extension of this work.

D. LIMITATIONS

All data derive from one database and one Bravais class, with 1,169 labelled materials. In the DFT-free tier the ensemble's advantage over a well-tuned XGBoost is not statistically significant. We compare against tabular baselines only, not against structure-aware graph networks [39]–[43], which require relaxed atomic coordinates — and which, we note,

would reintroduce exactly the DFT dependence the DFT-free tier is designed to avoid.

V. CONCLUSION

A physics-informed hybrid residual architecture, previously reported to improve band-gap prediction for two-dimensional materials, does not improve on the gradient-boosted prior it corrects once the evaluation is controlled for leakage, and none of its physics-motivated components contributes measurably. The earlier improvement is reproduced on demand by reverting a single implementation detail — building the tree prior in sample rather than out of fold — which trains the corrector to emit nothing and leaves a pass-through of the tree wearing the appearance of a hybrid. We regard this as the transferable finding: hybrid residual designs are widely reported to help, this particular failure is easy to introduce by accident, and it is invisible in the metrics such papers normally report.

Two positive results accompany it. A stacked ensemble of five heterogeneous learners is the strongest configuration tested, and we argue it should be justified by the risk it removes and the large errors it avoids at negligible absolute cost, rather than by a difference in R^2 that cannot carry the argument. Separately, decomposing accuracy by descriptor cost shows that composition and symmetry account for most of what is attainable here, with the full set of DFT energy channels adding 0.044 in R^2 .

Where a modest ensemble gain must be justified, we suggest it be justified as we have here: by the risk it removes and the errors it avoids, at a cost measured against the calculation it replaces, rather than by a difference in R^2 .

DATA AND CODE AVAILABILITY

All code, the curated dataset, per-fold out-of-sample predictions, and the scripts generating every table and figure are available at <https://github.com/adhvayjagadeesh/DFT--Machine-Learning--PINN>. Every experiment reproduces from a single command under a pinned environment with a fixed random seed.

REFERENCES

- [1] G. Fiori, F. Bonaccorso, G. Iannaccone, T. Palacios, D. Neumaier, A. Seabaugh, S. K. Banerjee, and L. Colombo, "Electronics based on two-dimensional materials," *Nature Nanotechnology*, vol. 9, no. 10, pp. 768–779, 2014.
- [2] J. R. Schaibley, H. Yu, G. Clark, P. Rivera, J. S. Ross, K. L. Seyler, W. Yao, and X. Xu, "Valleytronics in 2D materials," *Nature Reviews Materials*, vol. 1, p. 16055, 2016.
- [3] D. Xiao, G.-B. Liu, W. Feng, X. Xu, and W. Yao, "Coupled spin and valley physics in monolayers of MoS_2 and other group-VI dichalcogenides," *Physical Review Letters*, vol. 108, no. 19, p. 196802, 2012.
- [4] A. K. Singh, K. Mathew, H. L. Zhuang, and R. G. Hennig, "Computational screening of 2D materials for photocatalysis," *The Journal of Physical Chemistry Letters*, vol. 6, no. 6, pp. 1087–1098, 2015.
- [5] J. Heyd, G. E. Scuseria, and M. Ernzerhof, "Hybrid functionals based on a screened Coulomb potential," *The Journal of Chemical Physics*, vol. 118, no. 18, pp. 8207–8215, 2003.
- [6] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [7] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [8] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.
- [9] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022, pp. 507–520.
- [10] D. McElfresh, S. Khandagale, J. Valverde, G. Ramakrishnan, M. Goldblum, and C. White, "When do neural nets outperform boosted trees on tabular data?" in *Advances in Neural Information Processing Systems*, 2023.
- [11] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
- [12] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021.
- [13] A. S. Rodin, A. Carvalho, and A. H. Castro Neto, "Strain-induced gap modification in black phosphorus," *Physical Review Letters*, vol. 112, p. 176801, 2014.
- [14] L. Li, Y. Yu, G. J. Ye, Q. Ge, X. Ou, H. Wu, D. Feng, X. H. Chen, and Y. Zhang, "Black phosphorus field-effect transistors," *Nature Nanotechnology*, vol. 9, pp. 372–377, 2014.
- [15] F. Xia, H. Wang, and Y. Jia, "Rediscovering black phosphorus as an anisotropic layered material for optoelectronics and electronics," *Nature Communications*, vol. 5, p. 4458, 2014.
- [16] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.
- [17] S. Hastrup, M. Strange, M. Pandey, T. Deilmann, P. S. Schmidt, N. F. Hinsche, M. N. Gjerding, D. Torelli, P. M. Larsen, A. C. Riis-Jensen, J. Gath, K. W. Jacobsen, J. J. Mortensen, T. Olsen, and K. S. Thygesen, "The Computational 2D Materials Database: High-throughput calculations of two-dimensional materials," *2D Materials*, vol. 5, no. 4, p. 042002, 2018.
- [18] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, "A general-purpose machine learning framework for predicting properties of inorganic materials," *npj Computational Materials*, vol. 2, p. 16028, 2016.
- [19] L. Ward, A. Dunn, A. Faghaninia et al., "Matminer: An open source toolkit for materials data mining," *Computational Materials Science*, vol. 152, pp. 60–69, 2018.
- [20] Y. Zhang, W. Xu, G. Liu, Z. Zhang, J. Zhu, and M. Li, "Bandgap prediction of two-dimensional materials using machine learning," *PLOS ONE*, vol. 16, no. 8, p. e0255637, 2021.
- [21] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.
- [22] G. Varoquaux, "Cross-validation failure: Small sample sizes lead to large error bars," *NeuroImage*, vol. 180, pp. 68–77, 2018.
- [23] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [24] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik, "Support vector regression machines," in *Advances in Neural Information Processing Systems*, vol. 9, 1996, pp. 155–161.
- [25] S. Elfving, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural Networks*, vol. 107, pp. 3–11, 2018.
- [26] P. Ramachandran, B. Zoph, and Q. V. Le, "Searching for activation functions," *arXiv preprint arXiv:1710.05941*, 2017.
- [27] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proceedings of the International Conference on Machine Learning*. PMLR, 2015, pp. 448–456.
- [28] L. Breiman, "Stacked regressions," *Machine Learning*, vol. 24, no. 1, pp. 49–64, 1996.
- [29] R. Koenker and G. Bassett, "Regression quantiles," *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.
- [30] Y. Zhuo, A. Mansouri Tehrani, and J. Brgoch, "Predicting the band gaps of inorganic solids by machine learning," *The Journal of Physical Chemistry Letters*, vol. 9, no. 7, pp. 1668–1673, 2018.
- [31] L. C. Gomes et al., "The utility of composition-based machine learning models for band gap prediction," *Computational Materials Science*, vol. 197, p. 110995, 2021.
- [32] A. Zeb et al., "Machine learning-enhanced prediction of inorganic semiconductor bandgaps for advancing optoelectronic technologies," *Advanced Theory and Simulations*, vol. 7, p. 2400190, 2024.

- [33] Y. Wang *et al.*, “Accurate prediction of band gap of two-dimensional monolayer materials via transfer learning,” *Materials Today Communications*, 2025, 290 compositional descriptors over 2,915 C2DB monolayers.
- [34] T. Gneiting and A. E. Raftery, “Strictly proper scoring rules, prediction, and estimation,” *Journal of the American Statistical Association*, vol. 102, no. 477, pp. 359–378, 2007.
- [35] F. Tavazza, K. Choudhary, and B. DeCost, “Approaches for uncertainty quantification of AI-predicted material properties: A comparison,” *arXiv preprint arXiv:2310.13136*, 2023.
- [36] G. Airlangga and A. Liu, “A hybrid gradient boosting and neural network model for predicting urban happiness: Integrating ensemble learning with deep representation for enhanced accuracy,” *Machine Learning and Knowledge Extraction*, vol. 7, no. 1, p. 4, 2025.
- [37] V. Tran, R. Soklaski, Y. Liang, and L. Yang, “Layer-controlled band gap and anisotropic excitons in few-layer black phosphorus,” *Physical Review B*, vol. 89, p. 235319, 2014.
- [38] J. Kim, S. S. Baik, S. H. Ryu, Y. Sohn, S. Park, B.-G. Park, J. Denlinger, Y. Yi, H. J. Choi, and K. S. Kim, “Observation of tunable band gap and anisotropic Dirac semimetal state in black phosphorus,” *Science*, vol. 349, no. 6249, pp. 723–726, 2015.
- [39] T. Xie and J. C. Grossman, “Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties,” *Physical Review Letters*, vol. 120, no. 14, p. 145301, 2018.
- [40] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, “SchNet: A deep learning architecture for molecules and materials,” *The Journal of Chemical Physics*, vol. 148, no. 24, p. 241722, 2018.
- [41] C. Chen, W. Ye, Y. Zuo, C. Zheng, and S. P. Ong, “Graph networks as a universal machine learning framework for molecules and crystals,” *Chemistry of Materials*, vol. 31, no. 9, pp. 3564–3572, 2019.
- [42] K. Choudhary and B. DeCost, “Atomistic line graph neural network for improved materials property predictions,” *npj Computational Materials*, vol. 7, p. 185, 2021.
- [43] C. Chen and S. P. Ong, “A universal graph deep learning interatomic potential for the periodic table,” *Nature Computational Science*, vol. 2, no. 11, pp. 718–728, 2022.

ADHVAY JAGADEESH is a student researcher with the Aspiring Scholars Directed Research Program (ASDRP), Fremont, CA, USA. His research interests include machine learning for materials science, physics-informed neural networks, and computational modeling of two-dimensional materials.

RUTVI MUDALAGI is a high school student researcher with the Aspiring Scholars Directed Research Program (ASDRP), Fremont, CA, USA.

MARX AKL received the Ph.D. degree in physics. He is the faculty advisor with the Aspiring Scholars Directed Research Program (ASDRP), Fremont, CA, USA.

...